

Survival Analysis Project Report

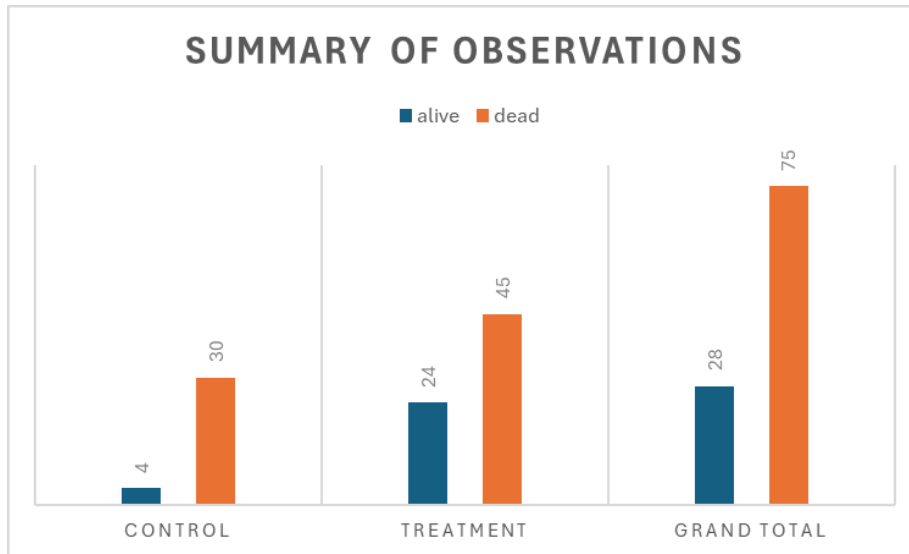
1- INTRODUCTION

The Stanford University Heart Transplant study was conducted from September 1967 till March 1974 to determine whether an experimental heart transplant program increased lifespan. The dataset consists of 103 observations on the following 9 variables : ID number of the patient , **acceptyear** (Year of acceptance as a heart transplant candidate) , **age** (Age of the patient at the beginning of the study) , **survived** (Survival status with levels alive and dead) , **survtime** (Number of days patients were alive after the date they were determined to be a candidate for a heart transplant until the termination date of the study), **prior** (Whether or not the patient had prior surgery with levels yes and no) , **transplant** (Transplant status with levels control (did not receive a transplant) and treatment (received a transplant)) , **wait** (Waiting Time for Transplant) and mismatch on antigen (measure of antigen incompatibility). Is the treatment effective? Are there any factors that influence the treatment? Our results suggest that not only receiving a transplant reduces the risk of death by 86.2% but being in the younger age group reduces the same risk by approximately 63.72%. Also, patients with a history of prior surgery have a 55% lower risk of death compared to patients without prior surgery.

2- Analysis

2-1 Summary of the data

The dataset consists of 103 men aged of 8 to 64. The mean age is 44.64-year-old. Out of them 75 Died or experienced the event: this is a 27.18 % censoring rate.



2-2 Significance of treatment

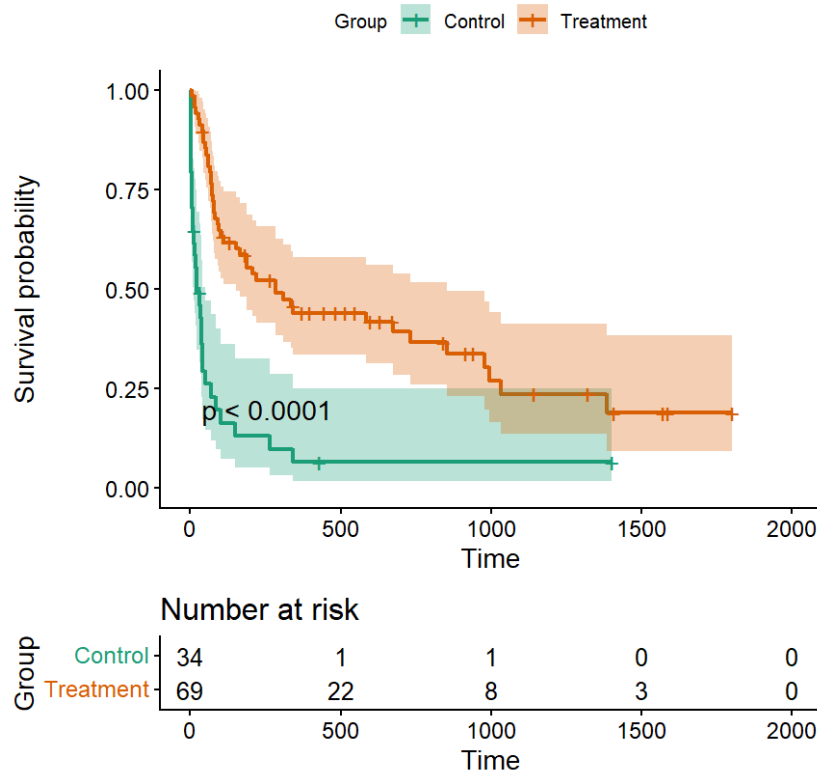
- LOG RANK TEST FOR TREATMENT

R Output

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
transplant=Control	34	30	12.1	26.5	33.2
transplant=Treatment	69	45	62.9	5.1	33.2

chisq= 33.2 on 1 degrees of freedom, p= 8e-09

P value = 8e-09 indicating that the two survival distributions are significantly different from one another



- COX PROPORTIONAL HAZARDS FOR TREATMENTS

Call:

```
coxph(formula = surv_object ~ transplant, data = data_clean)
```

n= 103, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant1	-1.3234	0.2662	0.2438	-5.428	5.69e-08 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant1	0.2662	3.756	0.1651	0.4293

Concordance= 0.668 (se = 0.025)

Likelihood ratio test= 25.95 on 1 df, p=4e-07

wald test = 29.47 on 1 df, p=6e-08

Score (logrank) test = 33.38 on 1 df, p=8e-09

Receiving a transplant reduces the risk of death by approximately 73.4% ($1 - 0.2662 = 0.7338$) compared to not receiving a transplant, and this effect is highly statistically significant.

2-3 Significance of age

- **Stratified Log-Rank Test on the transplant groups, using the age group as the stratification variable.**

Call:

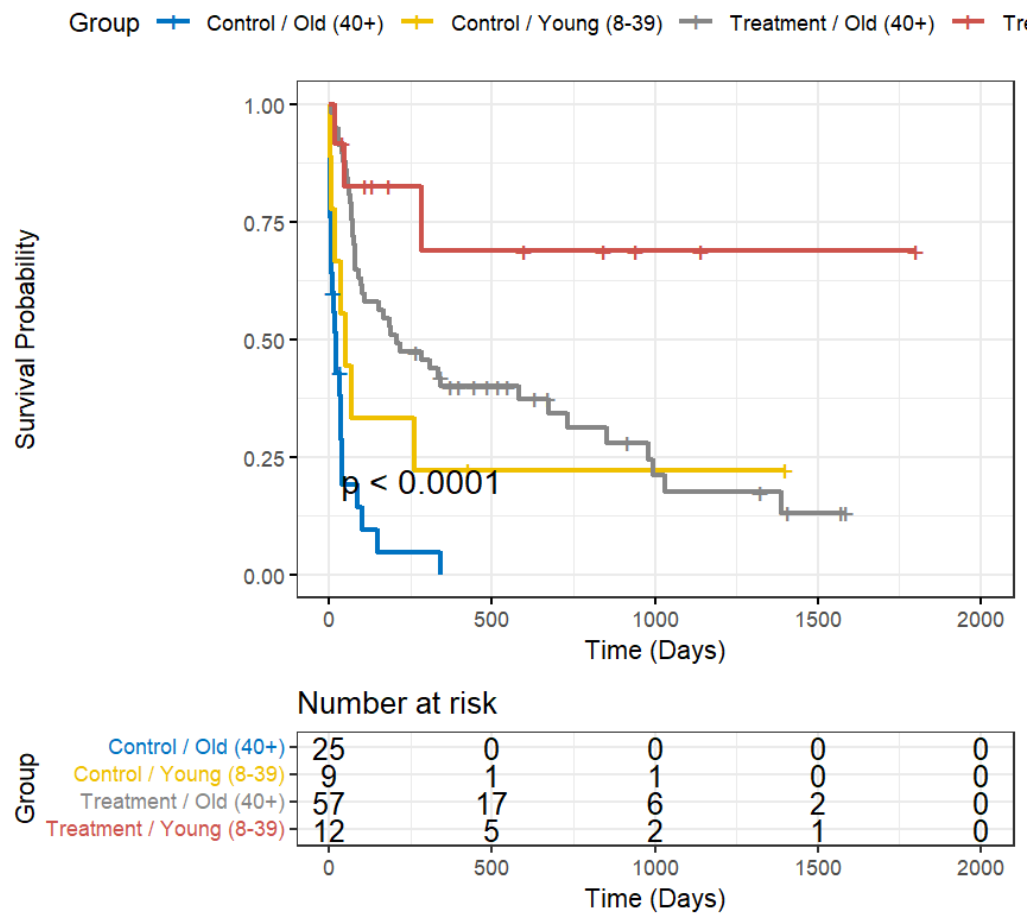
```
survdifff(formula = surv_object ~ combined_group, data = data_grouped)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
combined_group=0_40_plus	25	23	6.63	40.463	47.60
combined_group=0_8_39	9	7	5.47	0.431	0.47
combined_group=1_40_plus	57	42	51.45	1.736	5.65
combined_group=1_8_39	12	3	11.46	6.243	7.45

chisq= 52.4 on 3 degrees of freedom, p= 2e-11

The p-value shows us that we have a highly statistically significant difference in survival among the four groups.

Survival by Transplant Status and Age Group (FULL C



- Cox proportional hazards model of the effect of group age on survival

Call:

```
coxph(formula = surv_object ~ age_group, data = data_clean)
```

n= 103, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
age_group8_39	-0.6424	0.5260	0.3403	-1.888	0.059

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_group8_39	0.526	1.901	0.27	1.025

Concordance= 0.532 (se = 0.027)

Likelihood ratio test= 4.15 on 1 df, p=0.04

Wald test = 3.56 on 1 df, p=0.06

Score (logrank) test = 3.69 on 1 df, p=0.05

Effect: Being in the younger age group (8–39) is associated with a 47.4% reduction in the risk of death compared to the older group 40+

Significance: The p-value of 0.059 is just above 0.05 (the Likelihood Ratio test p=0.04 is significant). This suggests a weak-to-moderate association.

2-3 Significance of accepted year

- Stratified Log-Rank Test on the transplant groups, using the accepted year group as the stratification variable.

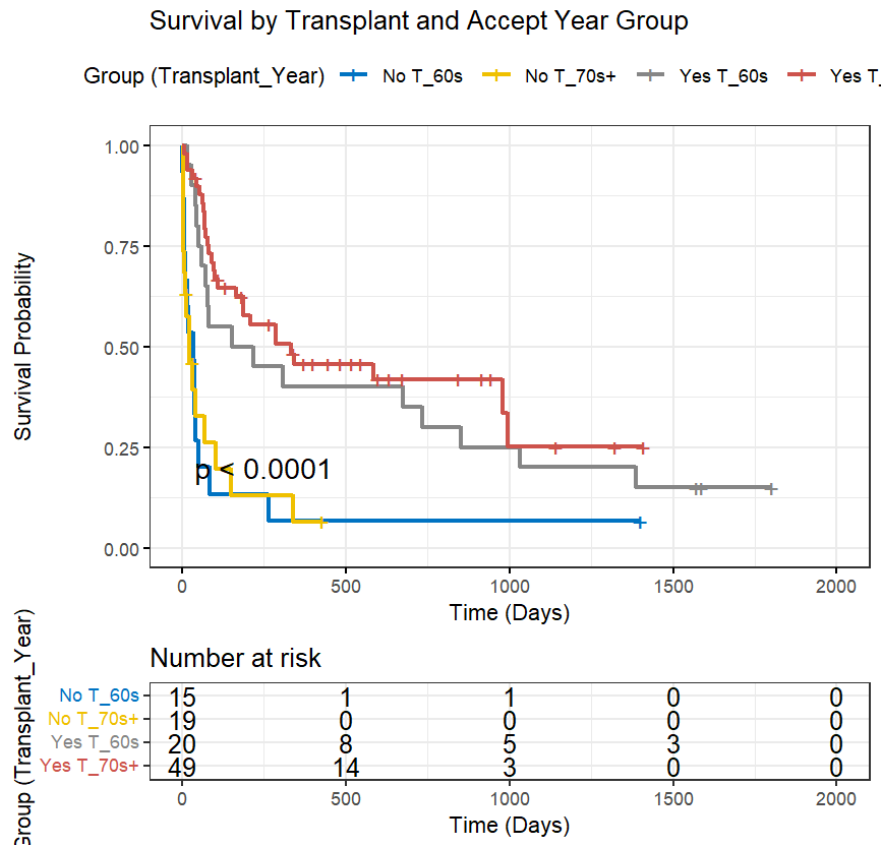
Call:

```
survdifff(formula = surv_object ~ combined_group, data = data_grouped)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
combined_group=0_60_69	15	14	5.82	11.515	12.82
combined_group=0_70_plus	19	16	6.27	15.073	17.06
combined_group=1_60_69	20	17	19.25	0.263	0.37
combined_group=1_70_plus	49	28	43.66	5.616	13.97

Chisq= 34.1 on 3 degrees of freedom, p= 2e-07

p= 2e-07 indicates a highly statistically significant difference in survival among the four combined groups. We conclude that the survival outcomes differ based on the combination of transplant status and acceptance year group.



- Cox PH Model of Accept Year Group

Call:

```
coxph(formula = surv_object ~ year_group, data = data_clean)
```

n= 103, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
year_group70_plus	-0.3524	0.7030	0.2390	-1.475	0.14

	exp(coef)	exp(-coef)	lower .95	upper .95
year_group70_plus	0.703	1.422	0.4401	1.123

Concordance= 0.548 (se = 0.03)

Likelihood ratio test= 2.12 on 1 df, p=0.1

wald test = 2.17 on 1 df, p=0.1

Score (logrank) test = 2.19 on 1 df, p=0.1

The P value of 0.14 is above 0.05. Therefore, we fail to reject the null hypothesis. The model is not statistically significant in predicting survival based on this grouping alone.

2-4 Full and Null Cox PH Model

- Fully Adjusted Cox Model

Call:

```
coxph(formula = surv_object ~ year_group + transplant + age_group,  
      data = data_final_adjusted)
```

n= 103, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
year_group70_plus	-0.2597	0.7713	0.2427	-1.070	0.284630
transplant1	-1.6421	0.1936	0.2646	-6.206	5.45e-10 ***
age_group8_39	-1.2165	0.2963	0.3635	-3.347	0.000818 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
year_group70_plus	0.7713	1.296	0.4794	1.2411
transplant1	0.1936	5.166	0.1152	0.3252
age_group8_39	0.2963	3.375	0.1453	0.6041

Concordance= 0.71 (se = 0.03)

Likelihood ratio test= 40.4 on 3 df, p=9e-09

Wald test = 41.76 on 3 df, p=5e-09

Score (logrank) test = 45.72 on 3 df, p=7e-10

Transplant is the Strongest Factor: Receiving a transplant reduces the hazard by 80.6% independent of the patient's age or acceptance year.

Age is a Major Independent Factor: Being in the younger age group reduces the hazard by 70.4%, independent of transplant status or acceptance year.

Acceptance Year has No Independent Effect: After accounting for the strong effects of transplant and age, the year group 70+ is not a significant predictor $p=0.284$. The apparent benefit seen in the Kaplan-Meier plot was fully explained by the better outcomes associated with transplant and younger age.

- Cox PH Null Model

```
Call: coxph(formula = surv_object ~ 1, data = data_final_adjusted)
```

Null model

log likelihood= -298.1214

n= 103

Significance of covariates

Model	logLikelihood	df	LRT	Conclusion
Null	-298.1214	0		
Transplant	-285.1483	1	25.9462	Significant
age	-296.046	1	4.1508	Significant
Acceptyear	-297.0615	1	2.1198	Not Significant

2-5 Better Model

- COX PROPORTION FOR TREATMENTS BY AGE GROUP

Call:

```
coxph(formula = surv_object ~ transplant + age_group, data = data_grouped)
```

n= 103, number of events= 75

```

              coef exp(coef) se(coef)      z Pr(>|z|)
transplant1 -1.6493    0.1922  0.2622 -6.290 3.18e-10 ***
age_group8_39 -1.1689    0.3107  0.3580 -3.265  0.00109 **
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

              exp(coef) exp(-coef) lower .95 upper .95
transplant1    0.1922      5.203    0.115    0.3213
age_group8_39  0.3107      3.218    0.154    0.6267
```

Concordance= 0.697 (se = 0.028)

Likelihood ratio test= 39.28 on 2 df, p=3e-09

Wald test = 41.77 on 2 df, p=9e-10

Score (logrank) test = 45.5 on 2 df, p=1e-10

Interpretation

Likelihood Ratio $p=3e-09$ Highly significant. The model with transplant and age_group is significant.

Transplant Effect Receiving a transplant reduces the risk of the event (death) by approximately 80.8%, when holding the age group constant. This is an extremely strong protective factor.

Age Group Effect Being in the younger age group reduces the instantaneous risk of the event by approximately 68.9%, when holding the transplant status constant. This confirms that age is a major, independent factor.

Conclusion: Both receiving a transplant and being in the younger age group significantly reduce the risk of the event.

- Model with Interaction Term

Call:

```
coxph(formula = surv_object ~ transplant * age_group, data = data_grouped)
```

n= 103, number of events= 75

```

              coef exp(coef) se(coef)      z Pr(>|z|)
transplant1    -1.65229   0.19161  0.28189 -5.861 4.59e-09 ***
age_group8_39   -1.17648   0.30836  0.44467 -2.646  0.00815 **
transplant1:age_group8_39  0.02145   1.02168  0.74364  0.029  0.97699
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

              exp(coef) exp(-coef) lower .95 upper .95
transplant1      0.1916      5.2189   0.1103   0.3329
age_group8_39    0.3084      3.2429   0.1290   0.7372
transplant1:age_group8_39  1.0217      0.9788   0.2379   4.3883

```

Concordance= 0.697 (se = 0.028)

Likelihood ratio test= 39.28 on 3 df, p=2e-08

Wald test = 41.89 on 3 df, p=4e-09

Score (logrank) test = 52.79 on 3 df, p=2e-11

P value = 0.97699. There is **no statistically significant interaction** between transplant status and age group.

- Confirmation of the better model

Model	AIC
Transplant	572.2967
Age	594.092
Acceptyear	596.123
Transplant, Age	560.9667
Full	561.842

The best fitting model from among these five, using the AIC criterion, is then the one using both the treatment and the age group as covariates.

- PH Assumption check

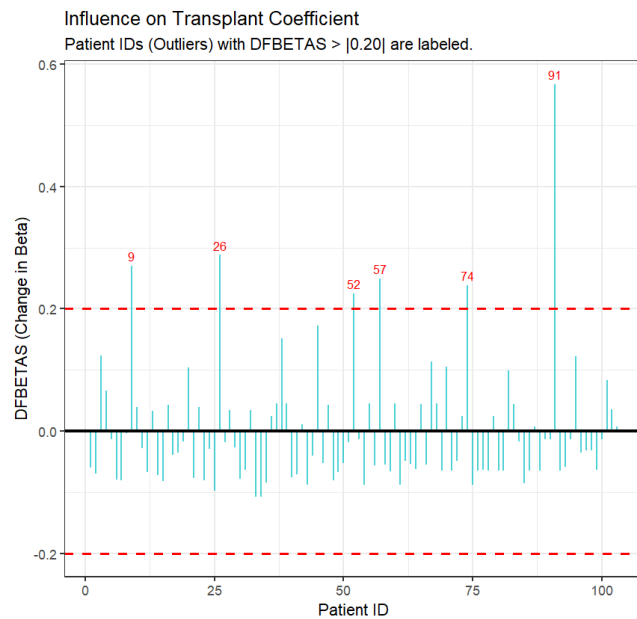
```
cox.zph(cox_model)
```

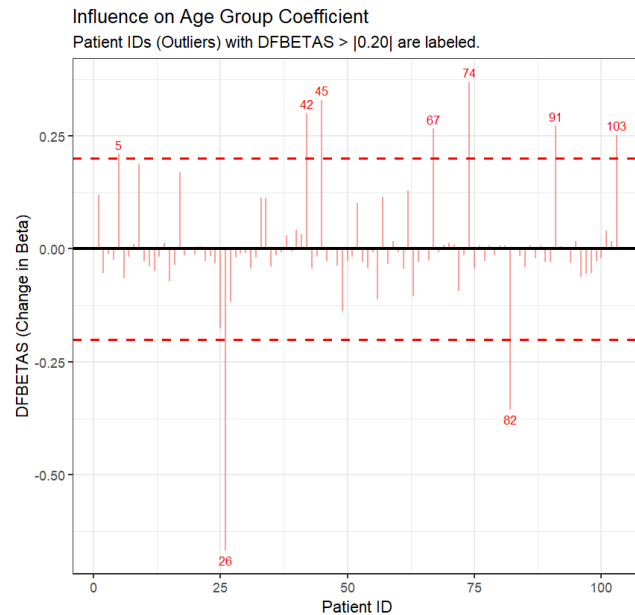
	chisq	df	p
transplant	5.77	1	0.016
age_group	3.87	1	0.049
GLOBAL	6.99	2	0.030

Since the p-values for both the main predictors are below 0.05, the assumption that the Hazard Ratios are constant over time is **rejected**. We must implement a **Time-Dependent Cox Model**

2-6 Model Diagnostics

- Case Deletion Residuals





Conclusion the model's stability is affected by 2 patients with unusually long event-free survival times (Patient ID 26 and 91).

- Sensitivity Analysis (Outliers Removed)

Call:

```
coxph(formula = surv_object_robust ~ transplant + age_group,
      data = data_robust)
```

n= 101, number of events= 74

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant1	-1.9808	0.1380	0.2821	-7.021	2.2e-12 ***
age_group8_39	-1.0138	0.3628	0.3536	-2.867	0.00414 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant1	0.1380	7.248	0.07937	0.2398
age_group8_39	0.3628	2.756	0.18143	0.7256

Concordance= 0.709 (se = 0.027)

Likelihood ratio test= 47.37 on 2 df, p=5e-11

wald test = 50.44 on 2 df, p=1e-11

Score (logrank) test = 60.18 on 2 df, p=9e-14

Transplant Effect: The protective effect of receiving a transplant became even stronger (0.1380 vs 0.1922). This suggests Patient ID91 (No Transplant, long survival) was artificially shrinking the measured benefit of the transplant group. When removed, the true protective effect is revealed to be over 86%

Age Group Effect: The protective effect of being in the younger age group became weaker (0.3628 vs 0.2963). This confirms Patient ID 26 (Young, extremely long event-free survival) was artificially inflating the protective effect of the younger age group.

Both transplant and age group remain highly significant and independent protective factors.

Proportion Hazards Assumption Check

```
> result.sch.resid
      chisq df    p
transplant  2.49  1 0.11
age_group   1.88  1 0.17
GLOBAL      3.46  2 0.18
```

Since all individual and global p-values are well above 0.05, we fail to reject the null hypothesis that the proportional hazards assumption is violated.

The Hazard Ratios previously computed are now considered stable and valid across the entire follow-up period. The Time-Dependent Cox Model (TDC) is no longer necessary. The simple additive Cox model is correct.

2-7 Factors that influence the treatment

- Weibull Model on Prior surgery and Mismatch on antigen

Call:

```
survreg(formula = surv_object_weibull ~ prior + Mismatch, data = data_transplant_analysis_clean,
        dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	6.278	0.259	24.22	<2e-16
prior1	1.204	0.639	1.88	0.0598
Mismatch1	-0.228	0.520	-0.44	0.6601
Log(scale)	0.366	0.122	2.99	0.0028

Scale= 1.44

weibull distribution

Loglik(model)= -326.8 Loglik(intercept only)= -329.1

Chisq= 4.58 on 2 degrees of freedom, p= 0.1

Number of Newton-Raphson Iterations: 5

n= 69

The overall model is **not statistically significant** (p=0.1). This suggests that prior surgery status and Mismatch status, *together*, do not significantly improve the prediction of survival time

- Cox Model on Prior surgery and Mismatch on antigen

```
coxph(formula = surv_object_final ~ prior + Mismatch, data = data_transplant_analysis_clean)
```

n= 69, number of events= 45

	coef	exp(coef)	se(coef)	z	Pr(> z)
prior1	-0.79907	0.44975	0.44283	-1.804	0.0712 .
Mismatch1	0.07245	1.07514	0.36738	0.197	0.8437

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
prior1	0.4497	2.2235	0.1888	1.071
Mismatch1	1.0751	0.9301	0.5233	2.209

Concordance= 0.571 (se = 0.037)

Likelihood ratio test= 3.95 on 2 df, p=0.1

wald test = 3.3 on 2 df, p=0.2

Score (logrank) test = 3.46 on 2 df, p=0.2

The predictors prior and Mismatch, together, do not significantly improve the model's prediction of survival

- Weibull on prior alone

```
Call:
survreg(formula = surv_object_weibull ~ prior, data = data_transplant_analysis_clean,
        dist = "weibull")

              Value Std. Error      z      p
(Intercept)  6.229      0.233 26.76 <2e-16
prior1       1.217      0.643  1.89  0.058
Log(scale)   0.374      0.121  3.09  0.002

Scale= 1.45

Weibull distribution
Loglik(model)= -326.9  Loglik(intercept only)= -329.1
      Chisq= 4.4 on 1 degrees of freedom, p= 0.036
Number of Newton-Raphson Iterations: 5
n= 69
```

The model is statistically significant $p=0.036$. This confirms that including prior surgery status significantly improves the prediction of survival time

- Cox Model for prior

```
Call:
coxph(formula = surv_object_final ~ prior, data = data_transplant_analysis_clean)

n= 69, number of events= 45

              coef exp(coef) se(coef)      z Pr(>|z|)
prior1 -0.7992    0.4497   0.4428 -1.805   0.0711 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
prior1    0.4497      2.224    0.1888    1.071

Concordance= 0.577 (se = 0.024 )
Likelihood ratio test= 3.91 on 1 df,  p=0.05
Wald test            = 3.26 on 1 df,  p=0.07
Score (logrank) test = 3.42 on 1 df,  p=0.06
```

The model is significant. The prior surgery status significantly improves the prediction of survival.

Effect: Patients with a history of prior surgery have a 55% lower risk of death compared to patients without prior surgery.

- Weibull on Mismatch alone

```
Call:
survreg(formula = surv_object_weibull ~ Mismatch, data = data_transplant_analysis_clean,
        dist = "weibull")
              Value Std. Error      z      p
(Intercept)  6.521      0.251 26.00 <2e-16
Mismatch1    -0.258      0.534 -0.48 0.6283
Log(scale)   0.393      0.124  3.17 0.0015
```

Scale= 1.48

Weibull distribution

Loglik(model)= -329 Loglik(intercept only)= -329.1

Chisq= 0.22 on 1 degrees of freedom, p= 0.64

Number of Newton-Raphson Iterations: 5

n= 69

The model is **not statistically significant** (p=0.64). This indicates that knowing a patient's Mismatch status does not significantly improve the model's ability to predict survival time

- Cox Model for Mismatch

Call:

```
coxph(formula = surv_object_final ~ Mismatch, data = data_transplant_analysis_clean)
```

n= 69, number of events= 45

	coef	exp(coef)	se(coef)	z	Pr(> z)
Mismatch1	0.07312	1.07586	0.36777	0.199	0.842

	exp(coef)	exp(-coef)	lower .95	upper .95
Mismatch1	1.076	0.9295	0.5232	2.212

Concordance= 0.498 (se = 0.034)

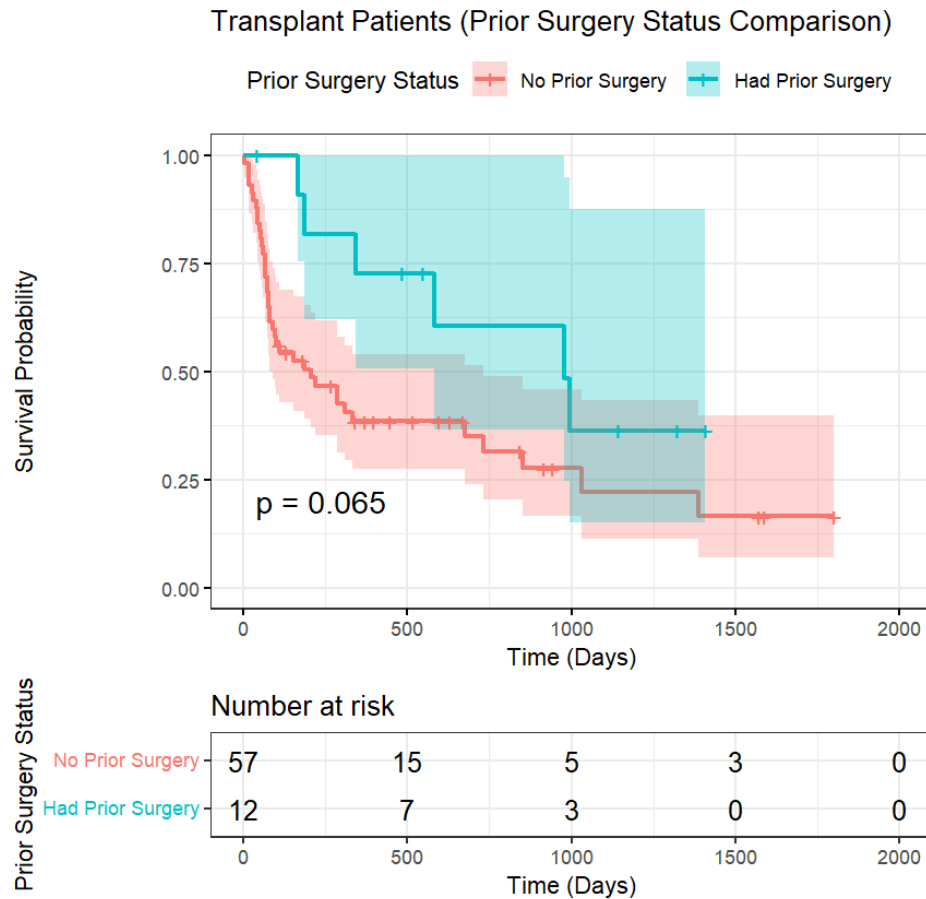
Likelihood ratio test= 0.04 on 1 df, p=0.8

wald test = 0.04 on 1 df, p=0.8

Score (logrank) test = 0.04 on 1 df, p=0.8

Not Significant. Mismatch status does not significantly improve the model's ability to predict survival.

Conclusion: Only prior surgery status can be used to predict survival.



3- CONCLUSION

We analyzed the Stanford Heart Transplant data collected from 1967 to 1974 to determine whether an experimental heart transplant program increased lifespan. Our results suggest that not only receiving a transplant reduces the risk of death by 86.2% but being in the younger age group reduces the same risk by approximately 63.72%. Furthermore, patients with a history of prior surgery have a 55% lower risk of death compared to patients without prior surgery.